Adebanji Oluwatimileyin Adelowo

Mathematical Engineer | Scientific Machine Learning Engineer

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Education

M.Sc. Mathematical Engineering | University of L'Aquila, Italy 2018 – 2021

- Thesis: *Lower Bounds on the Mix Norm of Passive Scalars Advected by Incompressible Enstrophy-Constrained Flows*
- Coursework: PDEs, Functional Analysis, Numerical Analysis, Finite Element Methods, Scientific Computing, Optimization, Machine Learning, Fluid Dynamics

B.Sc. Mathematics | Obafemi Awolowo University, Nigeria 2014 – 2018

- Thesis: *Application of Galerkin Finite Element Method to the One-Dimensional Stefan Problem*
- Coursework: Real Analysis, Complex Analysis, Topology, Numerical Analysis, PDEs, Probability, Mathematical Modeling

Research Experience

Master's Thesis Research | University of L'Aquila, Italy 2020 – 2021

- Derived analytical lower bounds on mix norms for passive scalars under energy and enstrophy constraints using variational methods and functional inequalities
- Identified optimal velocity fields accelerating mixing; implemented pseudo-spectral simulations in Python porting Gautam Iyer's MATLAB code

Undergraduate Thesis Research | Obafemi Awolowo University, Nigeria 2017 – 2018

- Applied Galerkin FEM to the one-dimensional Stefan (free-boundary) problem; developed algorithms for phase boundary tracking in Python

Professional Experience

Data Scientist | Intuos Srl, L'Aquila, Italy Dec 2024 – Present

- Developed ML Classifiers (Random Forest, LGBMClassifier) for real-time flight phase identification and FFT-based signal processing pipelines for engine RPM estimation and anomaly detection
- Built REST APIs, automated telemetry pipelines, and operational dashboards for flight performance analysis

Machine Learning Engineer | N-and Italia Srl, Pognano, Italy Apr 2023 – Nov 2024

- Developed predictive ML models for vending machine delivery failure forecasting from large-scale transaction and telemetry logs
- Built KPI dashboards and performed statistical analysis of system performance and operational anomalies

Operational Data Analyst | eResult Srl, Cesena, Italy Jan – Mar 2023

- Optimized SQL queries and automated reporting pipelines for business intelligence systems

Digital Engineering Intern | Bridgestone EMIA, Rome, Italy Apr – Oct 2022

- Improved deepcopy efficiency on GeometryMap half-edge data structures in tire cross-section FEA meshing pipelines
- Developed geometric algorithms for cubic Bézier curve closest-point computation using scalar minimization

Machine Learning Intern | Fellowship.AI, San Francisco (Remote) May – Aug 2022

- Implemented InsetGAN architecture for whole-body human image generation; deployed inference service to a Swift iOS app via FastAPI

Honors and Awards

University of L'Aquila International Scholarship | University of L'Aquila 2018 – 2021

Professor S.S. Okoya Prize | Excellence in Applied Mathematics, Obafemi Awolowo University 2018

MTN Foundation Scholarship | Merit-based undergraduate scholarship 2014 – 2018

PTDF Scholarship | Petroleum Technology Development Fund, Nigeria 2014 – 2018

Murli T. Chellaram Foundation Scholarship | Academic excellence award 2014 – 2018

Teaching and Mentorship

Founder & President, Inter-University Mathematics Club | Nigeria 2016 – 2018

- Established cross-university mentorship network; organized seminars on mathematics, data science, AI, and research careers

Academic Director, NAMSS | Peer Tutor, Obafemi Awolowo University, Nigeria 2015 – 2018

- Coordinated workshops and invited industry and academic speakers; tutored undergraduates in analysis and mathematical reasoning

Technical Skills

Programming: Python, SQL, MATLAB, C++, R, LaTeX

ML & AI: PyTorch, TensorFlow, scikit-learn, LangChain, OpenAI API

Engineering & Data: FastAPI, Docker, PostgreSQL, IBM DB2, Git, FFT, finite elements, pseudo-spectral methods

Visualization & Cloud: Matplotlib, Plotly, Power BI, React/Vite, AWS, Google Cloud

Selected Projects

INTUOS Flight Data Monitoring Dashboard | Full-stack aviation analytics platform; FastAPI + IBM DB2 backend, React/Vite frontend. Real-time telemetry, flight phase classification, engine metrics.

Aircraft Engine RPM Estimation via Audio DSP | Raspberry Pi system using FFT on microphone audio for RPM extraction; validated on Tecnam P92 / Rotax 912.

LLM Engineering Portfolio | Medical RAG Assistant, Enterprise NL-to-SQL, Financial Sentiment Distillation. Stack: LangChain, OpenAI, Pinecone.

Image Inpainting | TV-regularized Douglas–Rachford optimization for variational image restoration and inverse imaging.

Roof Segmentation | U-Net for aerial image segmentation with convolutional feature extraction and data augmentation.

Vending Machine Failure Prediction | Predictive ML for delivery failure forecasting from telemetry logs across 150+ machines.

Subsurface Skin Scattering | Dipole BSSRDF renderer for all 6 Fitzpatrick skin types in pure PyTorch; analytically derived and measured R_{hemi} across skin tones. *Stack: PyTorch, NumPy*

NeRF vs TensoRF | Pure-PyTorch NeRF and TensoRF trained on a synthetic scene; TensoRF achieves 17.2 dB PSNR / 0.834 SSIM vs NeRF's 14.2 dB / 0.770. *Stack: PyTorch, MPS*

Differentiable PBR Rendering | Cook–Torrance differentiable renderer optimising albedo, surface normals, roughness, and metallic maps from multi-view observations via photometric loss. *Stack: PyTorch, NumPy*

3D Liver Segmentation — Abdominal CT | 3D U-Net trained end-to-end on MSD Task03 liver CT volumes (131 cases). Foreground-biased patch sampling, combined soft-Dice + BCE loss, cosine LR annealing, Gaussian sliding-window inference. Best val Dice **0.9886**, HD95 $\leq$1 mm after 200 epochs on Kaggle T4. *Stack: PyTorch, MONAI, NumPy*